Digital Systems Design Lab.

Lecture 4

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Digital Systems Design Lab
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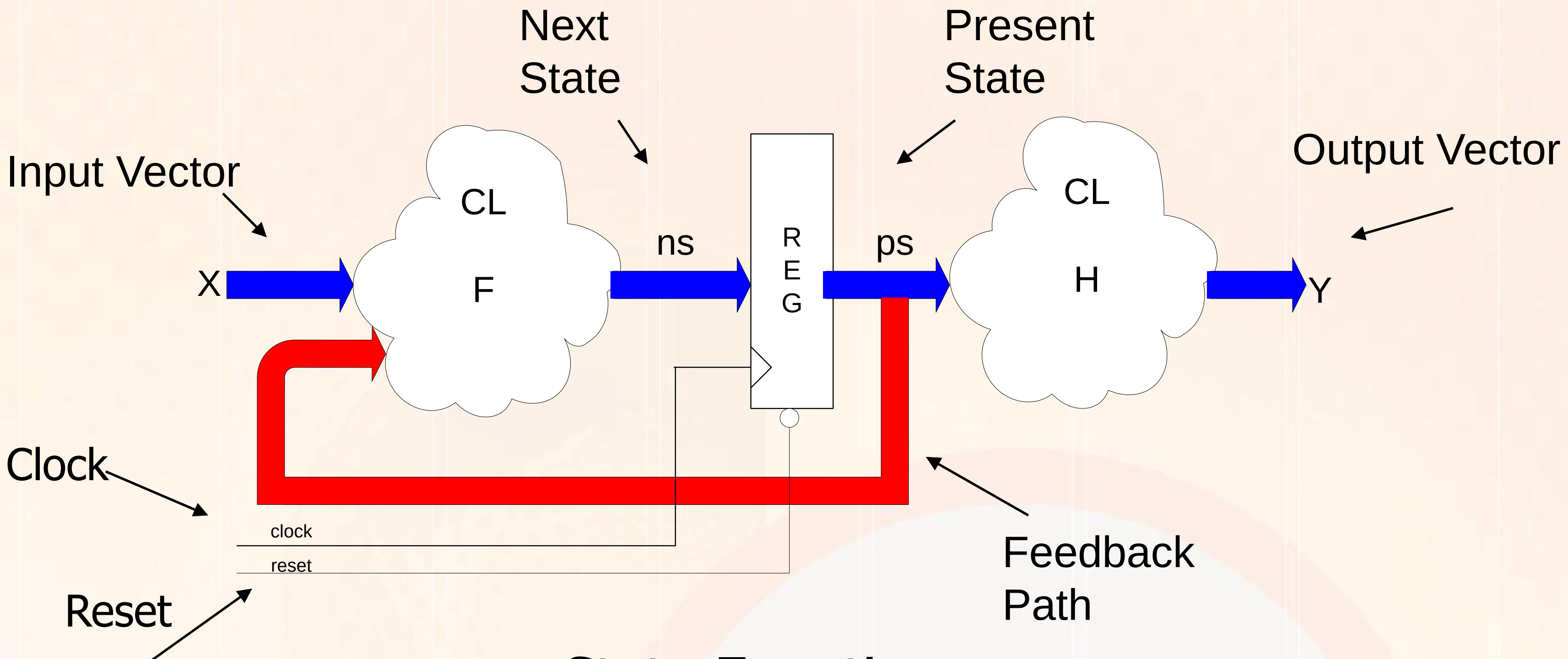
Mealy and Moore

- *Sequential machines are typically classified as either a Mealy machine or a Moore machine implementation.*
- *Moore machine: The outputs of the circuit depend only upon the present state of the circuit.*
- *Mealy machine: The outputs of the circuit depend upon both the present state of the circuit and the inputs.*

Moore Sequential Circuits



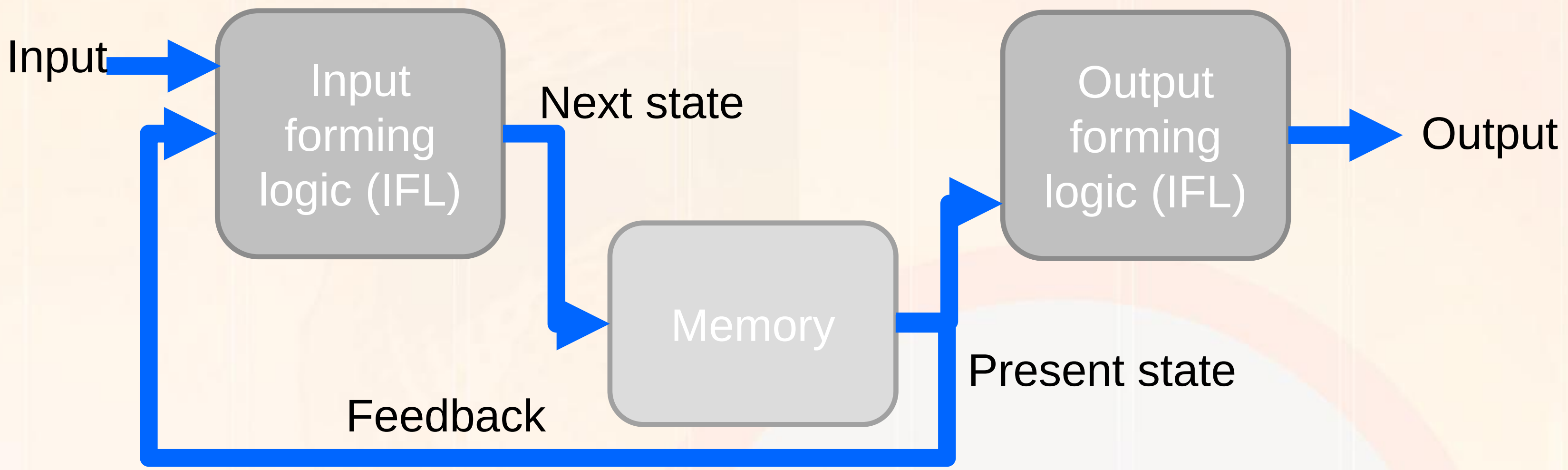
State Equations

$$n_s = F\left(X, p_s\right)$$

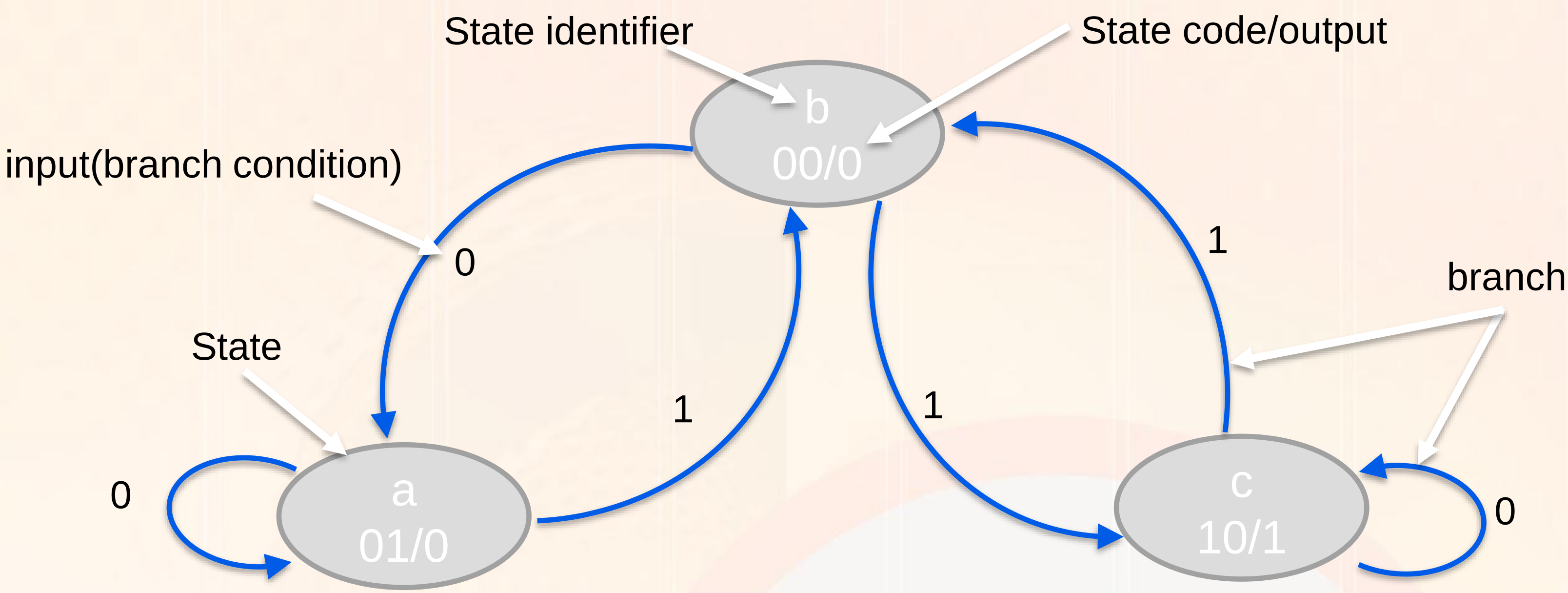
$$Y = H\left(p_s\right)$$

Moor model

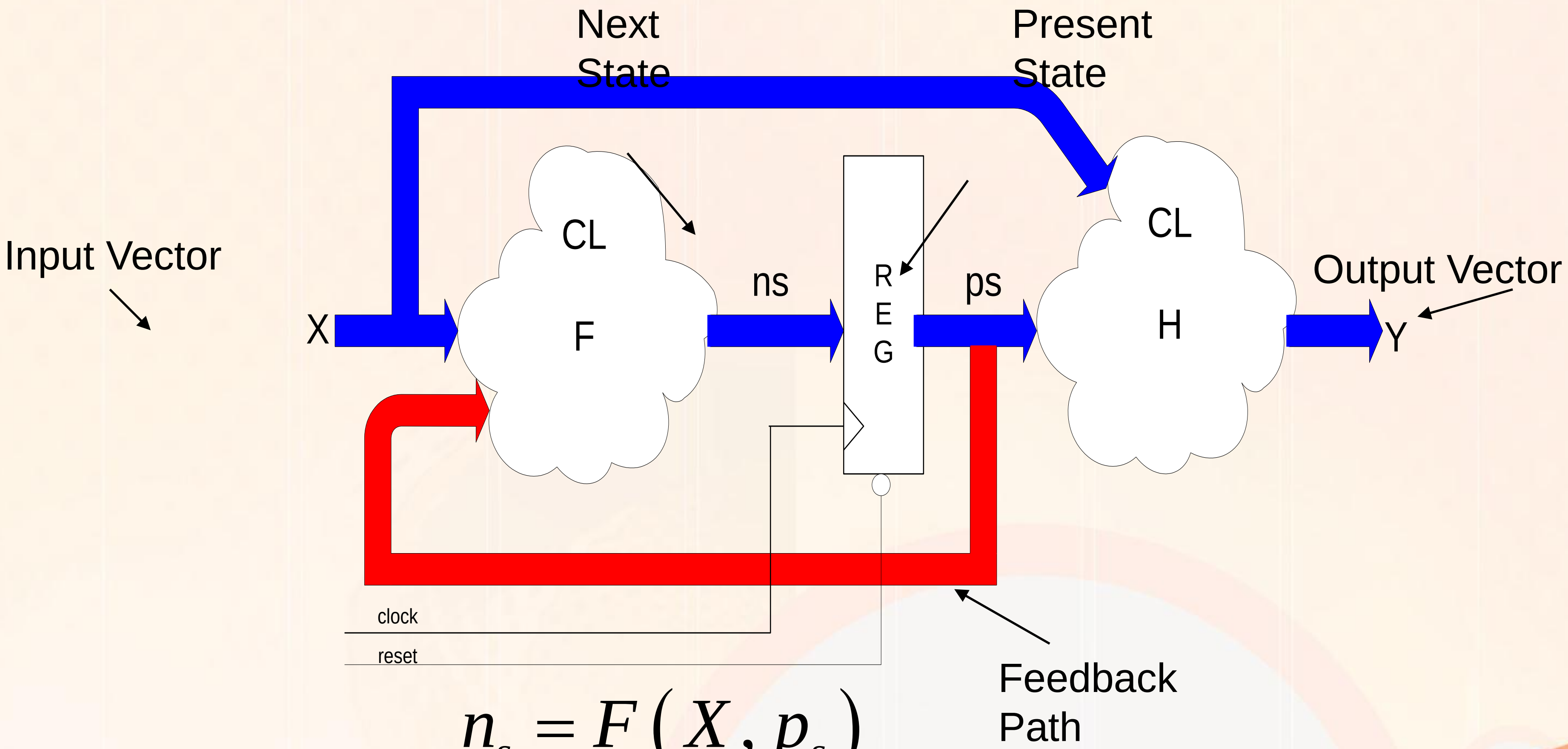
- *In a Moore model outputs of a circuit are function of the present state only.*



State Diagram/ Moore: Example



Mealy Sequential Circuits



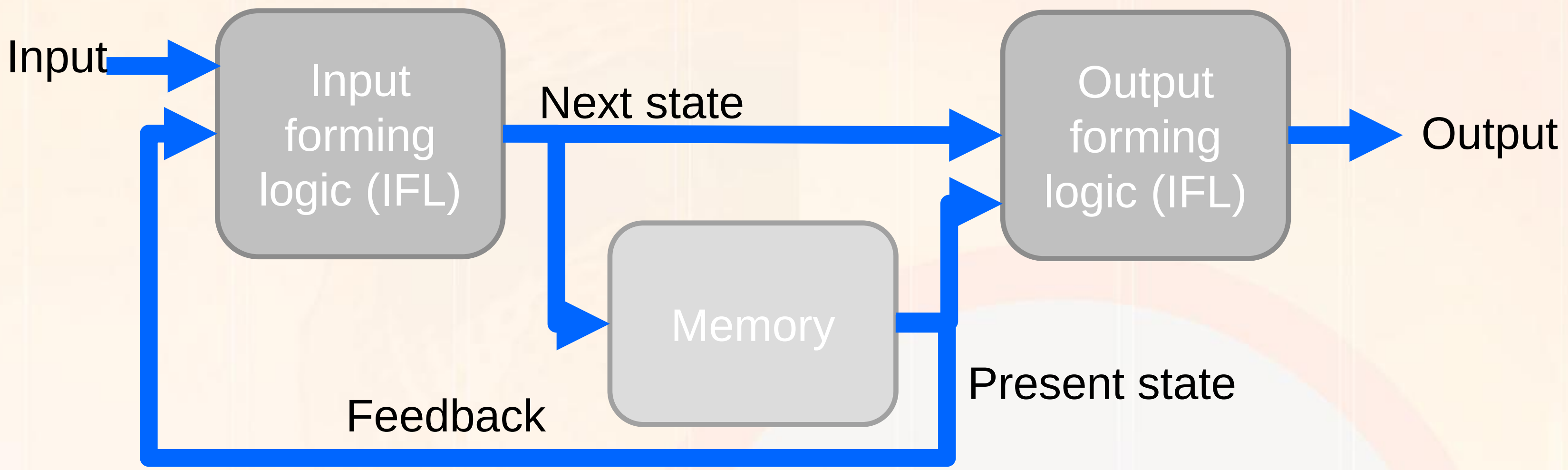
$$n_s = F\left(X, p_s\right)$$

$$Y = H\left(X, p_s\right)$$

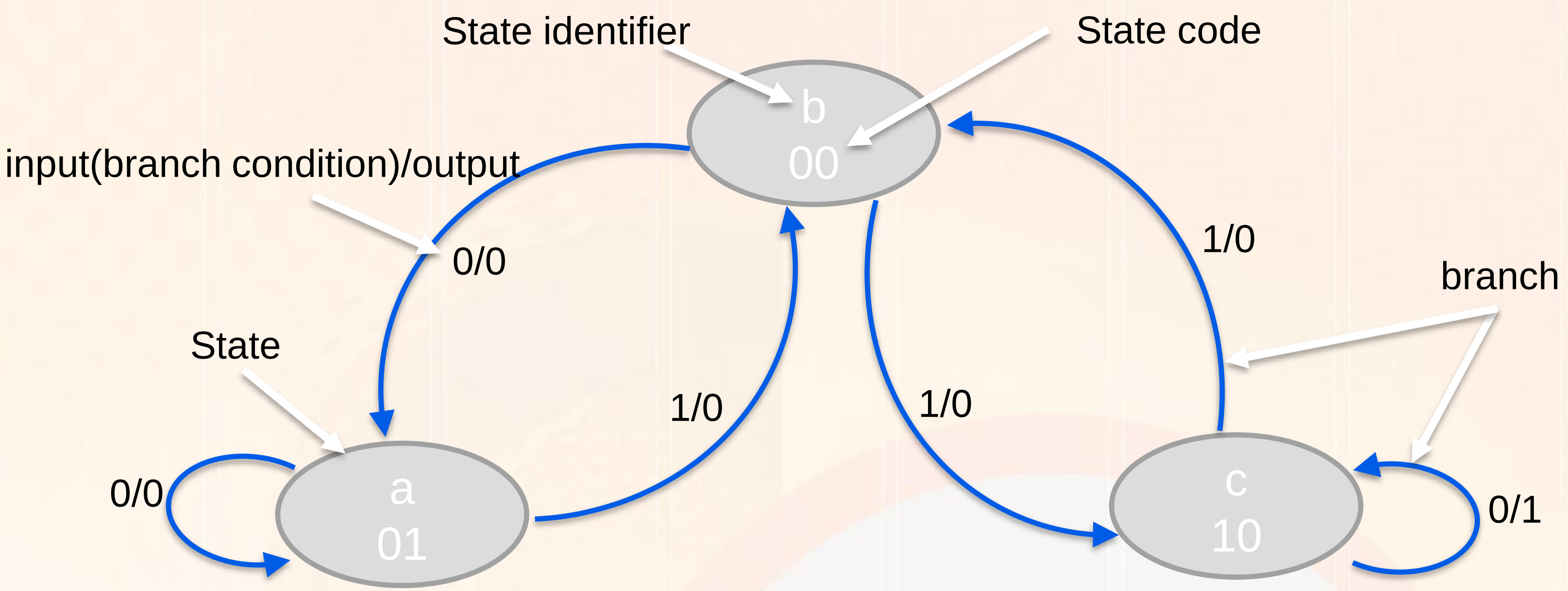
Output Y is also a function of input X

Mealy model

- *In a Mealy model outputs of a circuit are a function of the present state and the present inputs of the circuit.*



State Diagram/ Mealy : Example

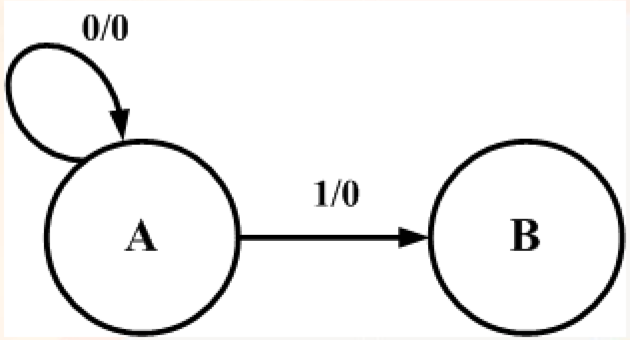


An example to go through the steps

- *The specification: The circuit will have one input, X, and one output, Z. The output Z will be 0 except when the input sequence 1101 are the last 4 inputs received on X. In that case it will be a 1.*

Generation of a state diagram

- *Create states and meaning for them.*
  - *State A – the last input was a 0 and previous inputs unknown. Can also be the reset state.*
  - *State B – the last input was a 1 and the previous input was a 0. The start of a new sequence possibly.*
- *Capture this in a state diagram*



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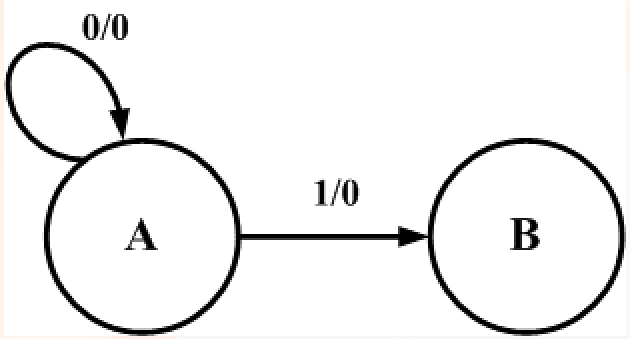
Notes on State diagrams

- *Capture this in a state diagram*
  - *Circles represent the states*
  - *Lines and arcs represent the transition between state.*
  - *The notation Input/Output on the line or arc specifies the input that causes this transition and the output for this change of state.*



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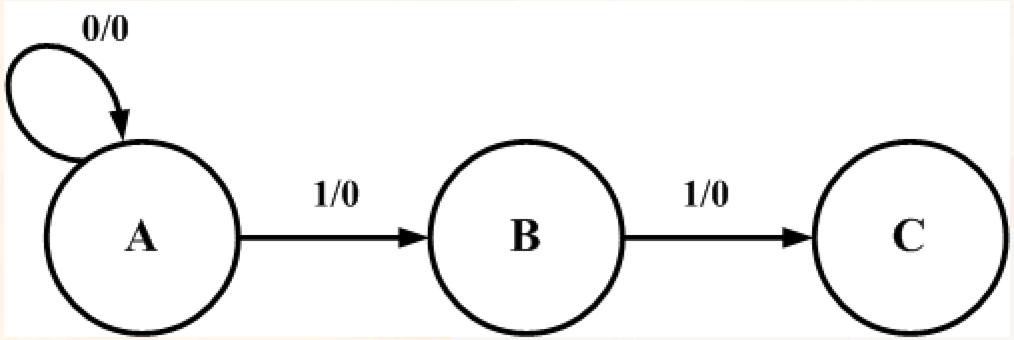
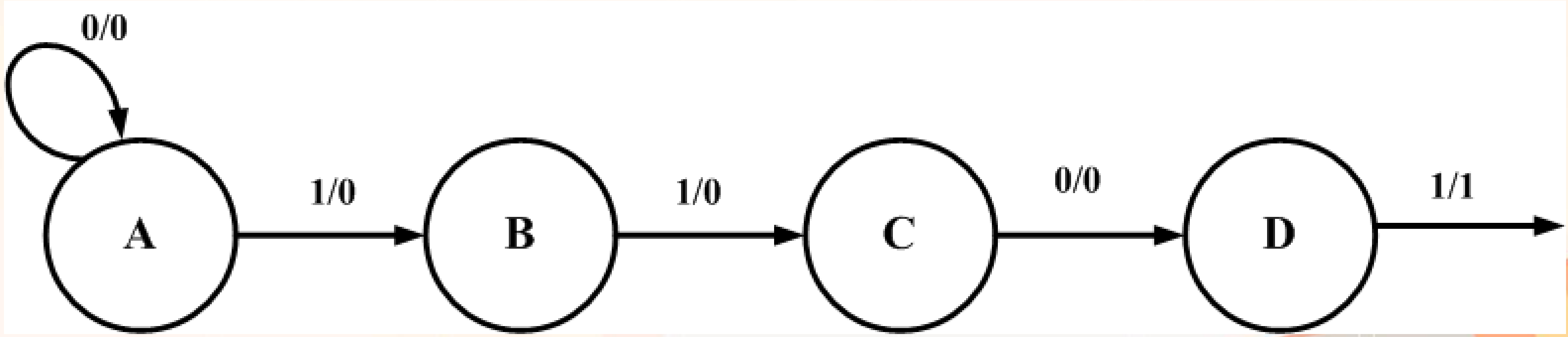
Continue to build up the diagram

- *Add a state C*
  - *State C – Have detected the input sequence 11 which is the start of the sequence.*



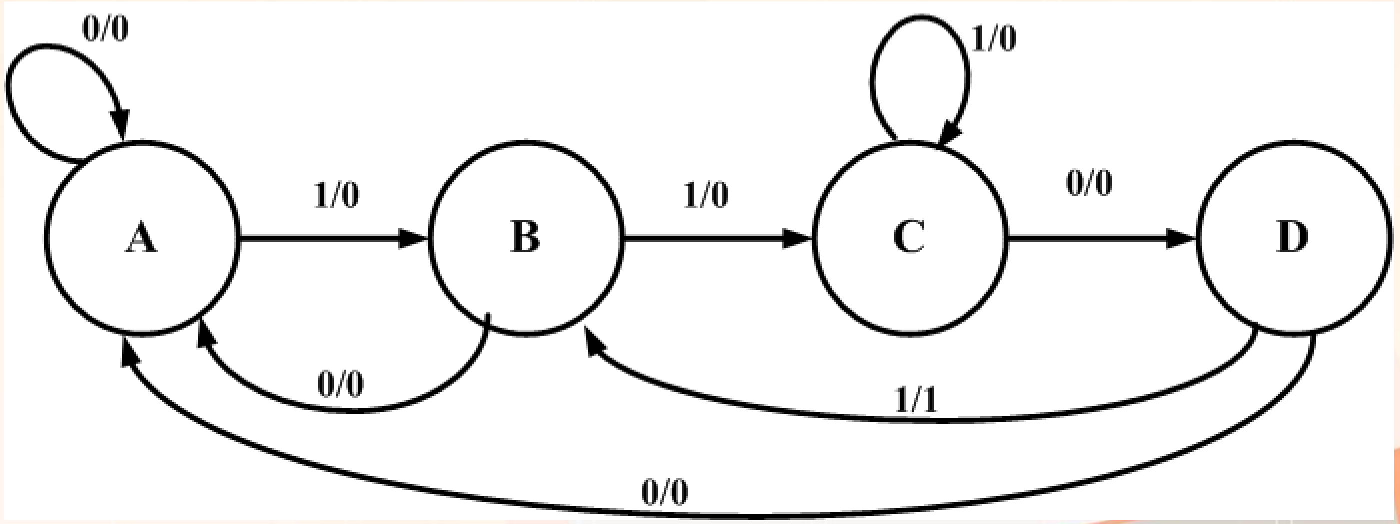
Continue

- *Add a state D*
  - *State D – have detected the 3rd input in the start of a sequence, a 0, now having 110. From State D, if the next input is a 1 the sequence has been detected and a 1 is output.*



Add remaining transitions

- *The previous diagram was incomplete.*
- *In each state the next input could be a 0 or a 1. This must be included.1101*



Now generate a state table

- *The state table*
- *This can be done directly from the state diagram.*

Prresent State	Next State X =0	Next State X=1	Output X=0	Output X=1
A	A	B	0	0
B	A	C	0	0
C	D	C	0	0
D	A	B	0	1

- *Now need to do a state assignment*

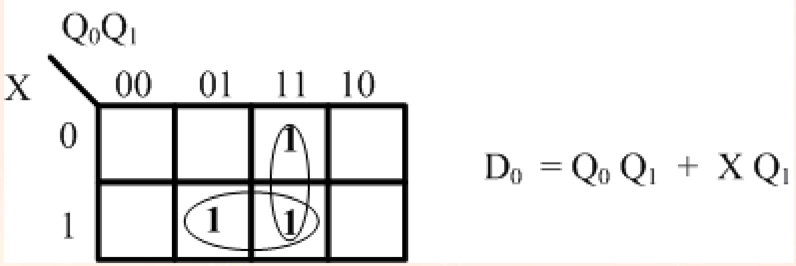
Select a state assignment

- *Will select a gray encoding*
- *For this state A will be encoded 00, state B 01, state C 11 and state D 10*

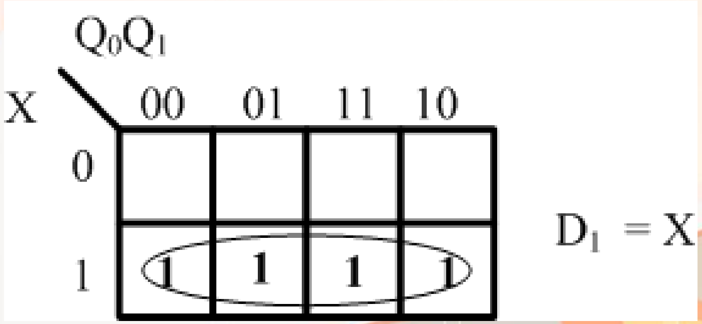
Present State	Next State X=0	Next State X=1	Output X=0	Output X=1
00	00	01	0	0
01	00	11	0	0
11	10	11	0	0
10	00	01	0	1

Flip-flop input equations

- *Generate the equations for the flip-flop inputs*
- *Generate the D_0 equation*



- *Generate the D_1 equation*



The output equation

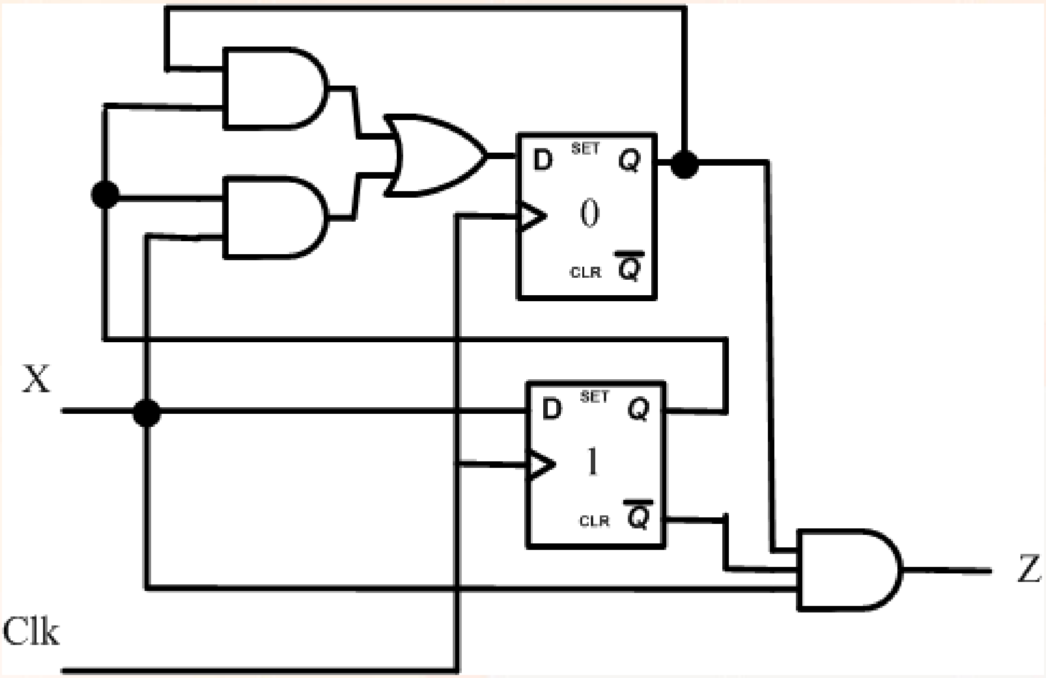
- *The next step is to generate the equation for the output Z and what is needed to generate it.*
- *Create a K-map from the truth table.*

X \ Q_0Q_1	00	01	11	10
0				
1				1

$$Z = X\,Q_0\,\overline{Q_1}$$

Now map to a circuit

- *The circuit has 2 D type F/Fs*



State Table: Mealy Example

- *2 bit in state encoding $\longrightarrow$ 2 Flipflops*
- *Determine the flip-flop excitation and sequential circuit output logic equations.*

$$Q_0^* = D_0 = ?$$

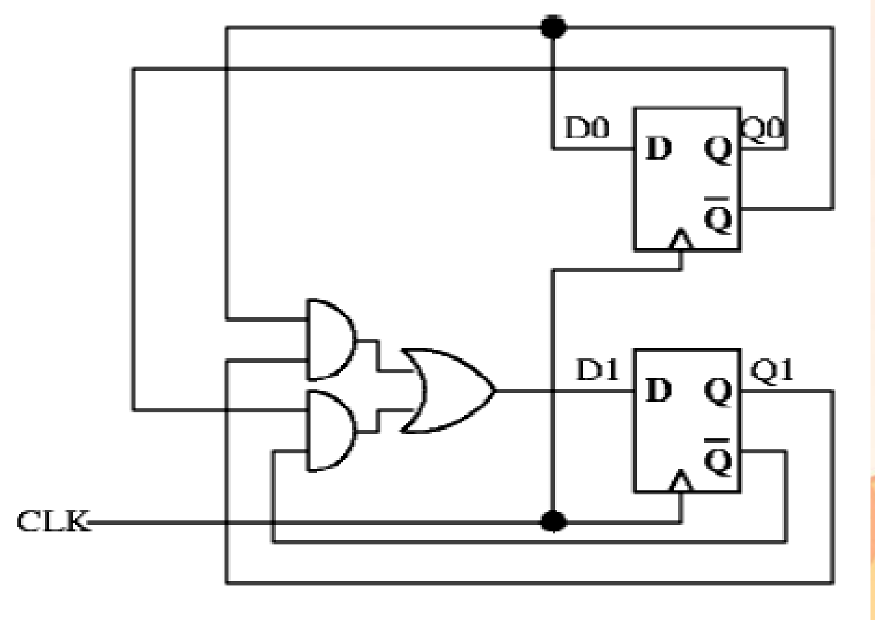
$$Q_1^* = D_1 = ?$$

input	Current state		Next state		output
X	Q_0	Q_1	Q_0^*	Q_1^*	Z
0	0	0	0	1	0
0	0	1	0	1	0
0	1	0	1	0	1
0	1	1	X	X	X
1	0	0	1	0	0
1	0	1	0	0	0
1	1	0	0	0	0
1	1	1	X	X	X

Example 2:

- *Determine the flip-flop excitation and sequential circuit output logic equations. (there are no explicit outputs in the above circuit).*

$$D_0 = \overline{Q_0}$$
$$D_1 = \overline{Q_0}Q_1 + Q_0\overline{Q_1}$$



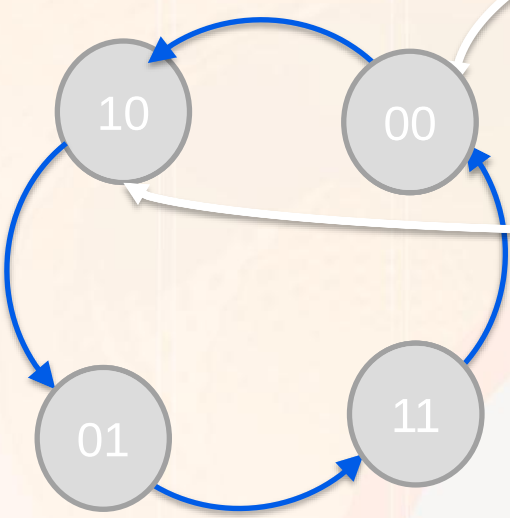
Example2:

$$D_0 = \overline{Q_0}$$

$$D_1 = \overline{Q_0}Q_1 + Q_0\overline{Q_1}$$

State table

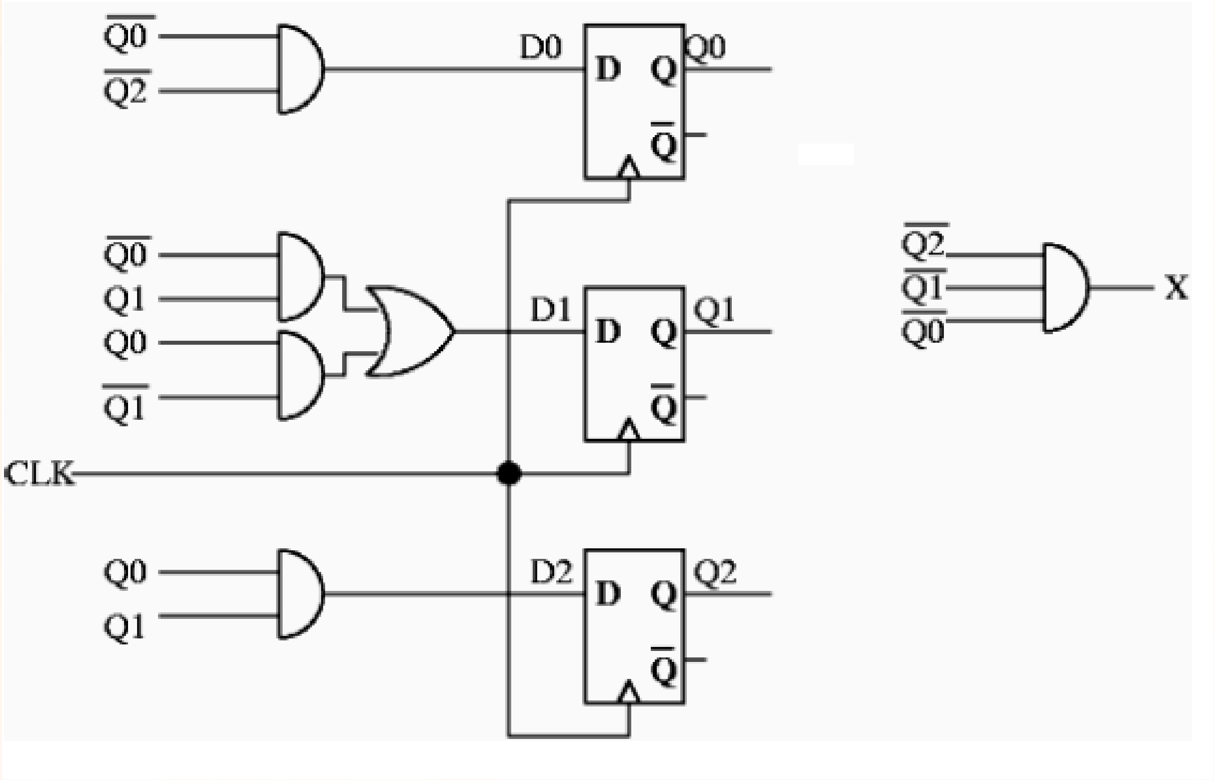
input	Current state		Next state		Output
none	Q_0	Q_1	Q_0^*	Q_1^*	None
	0	0	1	0	
	0	1	1	1	
	1	0	0	1	
	1	1	0	0	



State Diagram

Example 3

- *Determine the state diagram and the state table?*



Example 3

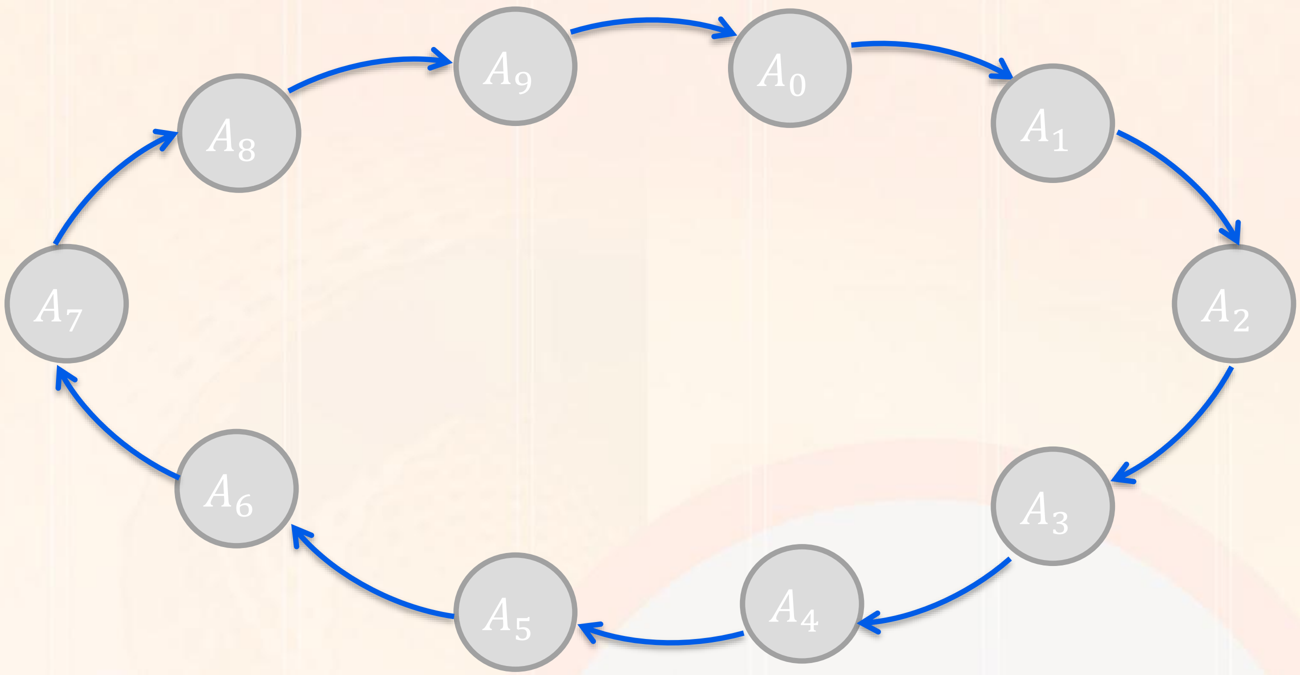
2. **State table**:

$$D_0 = \overline{Q_2}\,\overline{Q_0}\,; \qquad D_1 = Q_1\overline{Q_0} + \overline{Q_1}\,Q_0\,; \quad D_2 = Q_1 Q_0\,;$$

$$X = \overline{Q_2}\,\overline{Q_1}\,\overline{Q_0}$$

Input (*none*)	Present State			Flip-flop Inputs			Next State			Output
	Q_2	Q_1	Q_0	D_2	D_1	D_0	Q_2^*	Q_1^*	Q_0^*	X
	0	0	0	0	0	1	0	0	1	1
	0	0	1	0	1	0	0	1	0	0
	0	1	0	0	1	1	0	1	1	0
	0	1	1	1	0	0	1	0	0	0
	1	0	0	0	0	0	0	0	0	0
	1	0	1	0	1	0	0	1	0	0
	1	1	0	0	1	0	0	1	0	0
	1	1	1	1	0	0	1	0	0	0

Design Ex 1: Design a counter that count from 0 to 9?



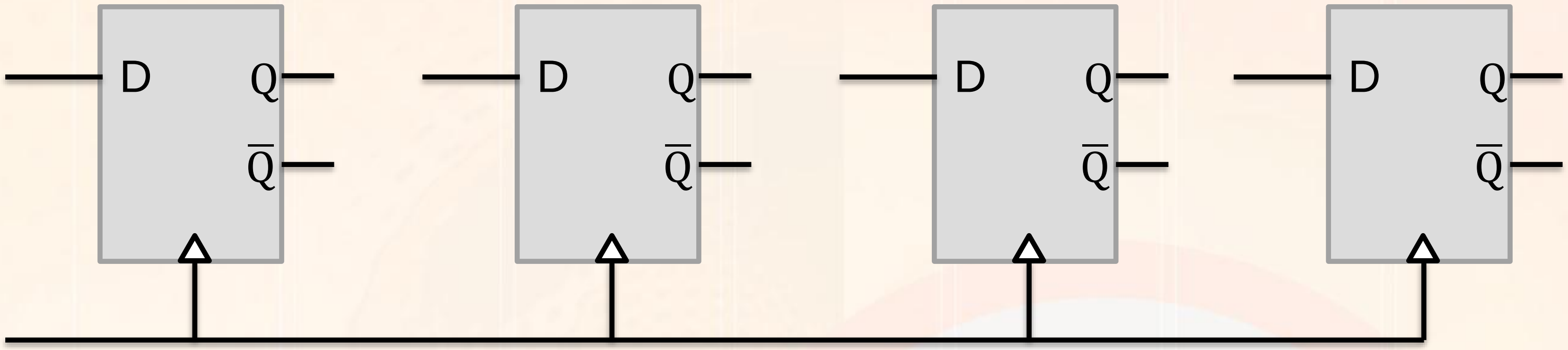
Design Example 1

- *Design a counter that count from 0 to 9?*

Current state				Next state			
Q_0	Q_1	Q_2	Q_3	Q_0^*	Q_1^*	Q_2^*	Q_3^*
0	0	0	0	0	0	0	1
0	0	0	1	0	0	1	0
0	0	1	0	0	0	1	1
0	0	1	1	0	1	0	0
0	1	0	0	0	1	0	1
0	1	0	1	0	1	1	0
0	1	1	0	0	1	1	1
0	1	1	1	1	0	0	0
1	0	0	0	1	0	0	1
1	0	0	1	0	0	0	0

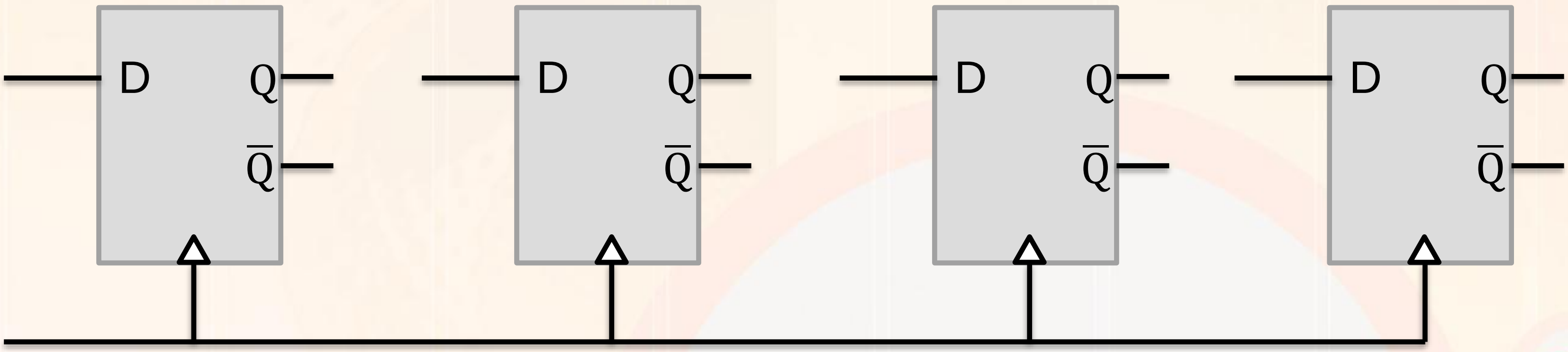
Design Example 1

- *4 bit coding 4 d-flip flops*



Design Example 1

$$D_0 = Q_1 Q_2 Q_3 + Q_0 \overline{Q_2}\,\overline{Q_3}$$
$$D_1 = Q_1 \overline{Q_2} + \overline{Q_1} Q_2 Q_3 + Q_1 Q_2 \overline{Q_3}$$
$$D_2 = Q_1 Q_2 Q_3 + Q_0 \overline{Q_2}\,\overline{Q_3}$$
$$D_3 = Q_2 \overline{Q_3} + \overline{Q_2}\,\overline{Q_3}$$



Design Example 1